Linear Algebra for Group Representations

Week 4 · Lecture 7

Mehmet Özkan

Group Theory

Reading for Lecture 7

- **Zee, linear-algebra review, pp. 1–8:** matrices, products, inverses and trace.
- **Zee, pp. 19–25:** adjoints, eigenvectors, diagonalization, inner products and unitary matrices.
- **Bünemann, §4.1, pp. 45–49:** matrix groups, equivalence and invariant subspaces.

Vectors are columns. Products act from right to left. We begin over $\mathbb{R}$ and introduce $\mathbb{C}$ before Lecture 8. All sums are written explicitly.

Group Theory

Vector spaces

A vector space V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ allows vector addition and scalar multiplication. For $u, v, w \in V$ and $a, b \in \mathbb{F}$, these operations stay in V and satisfy

$$u + v = v + u, \qquad (u + v) + w = u + (v + w),$$
$$v + 0 = v, \qquad v + (-v) = 0,$$
$$a(u + v) = au + av, \qquad (a + b)v = av + bv,$$
$$a(bv) = (ab)v, \qquad 1v = v.$$

Examples are $\mathbb{R}^n$, $\mathbb{C}^n$, and polynomials of degree at most two. Under addition, a vector space is an Abelian group; scalar multiplication is extra structure.

Group Theory

Subspaces

A subset $W \subseteq V$ is a subspace if it contains 0 and is closed under linear combinations:

$$u, v \in W, \quad a, b \in \mathbb{F} \quad \Longrightarrow \quad au + bv \in W.$$

For example, $W = \{(x, y)^T : y = 2x\}$ is a subspace of $\mathbb{R}^2$:

$$a \begin{pmatrix} x \\ 2x \end{pmatrix} + b \begin{pmatrix} z \\ 2z \end{pmatrix} = \begin{pmatrix} ax + bz \\ 2(ax + bz) \end{pmatrix} \in W.$$

The line $y = 2x+1$ is not a subspace: it does not contain the zero vector. A subspace inherits its addition and scalar multiplication from V.

Group Theory

Linear combinations and span

A linear combination of $v_1, \ldots, v_k$ is $a_1 v_1 + \cdots + a_k v_k$. Their span is the set of all such combinations:

$$\operatorname{span}\{v_1, \ldots, v_k\} = \left\{ \sum_{j=1}^{k} a_j v_j : a_j \in \mathbb{F} \right\}.$$

For $e_1 = (1, 0)^T$ and $e_2 = (0, 1)^T$,

$$\begin{pmatrix} x \\ y \end{pmatrix} = x e_1 + y e_2, \qquad \operatorname{span}\{e_1, e_2\} = \mathbb{R}^2.$$

By contrast, $\operatorname{span}\{(1, 2)^T\}$ is only the line $y = 2x$. Spanning means that every vector in the space can be obtained.

Group Theory

Linear independence

Vectors $v_1, \ldots, v_k$ are linearly independent if

$$a_1 v_1 + \cdots + a_k v_k = 0 \quad \Longrightarrow \quad a_1 = \cdots = a_k = 0.$$

For example, $(1, 2)^T$ and $(2, 4)^T$ are dependent because twice the first minus the second vanishes.

For $f_1 = (1, 1)^T$ and $f_2 = (1, -1)^T$,

$$a f_1 + b f_2 = 0 \quad \Longrightarrow \quad a + b = 0, \quad a - b = 0.$$

Adding and subtracting give $a = b = 0$. These two vectors are independent.

Group Theory

Bases and coordinates

An ordered basis is an independent list that spans the space. Every vector then has a unique expression

$$v = \sum_{j=1}^{n} x_j e_j.$$

Why unique? If $\sum_j x_j e_j = \sum_j y_j e_j$, independence applied to $\sum_j (x_j - y_j) e_j = 0$ gives $x_j = y_j$ for every j.

The numbers x_j are the coordinates of v in this basis. The number of basis vectors is the dimension. Thus $\dim \mathbb{R}^2 = 2$, while the polynomial space with basis $(1, x, x^2)$ has dimension three.

Group Theory

The same vector in two bases

Take $f_1 = (1,1)^T$, $f_2 = (1,-1)^T$. To express $(x,y)^T$ in this basis, solve

$$\begin{pmatrix} x \\ y \end{pmatrix} = a f_1 + b f_2 = \begin{pmatrix} a+b \\ a-b \end{pmatrix}.$$

Hence $a = (x+y)/2$ and $b = (x-y)/2$. This also proves that f_1, f_2 span $\mathbb{R}^2$, so they form a basis.

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2f_1 + f_2, \qquad [v]_{\mathcal{E}} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad [v]_{\mathcal{F}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

The vector is unchanged. Its coordinate column depends on the basis.

Group Theory

Linear maps

A map $T : V \to W$ is linear if

$$T(au + bv) = aT(u) + bT(v).$$

It preserves both addition and scalar multiplication. In particular, $T(0) = 0$.

The reflection $T(x, y) = (y, x)$ is linear. A translation $T(x) = x + a$, with fixed $a \neq 0$, is not: it sends 0 to a. Nor is $T(x, y) = (x^2, y)$ linear, since

$$T(2, 0) = (4, 0) \neq 2T(1, 0) = (2, 0).$$

Textbook: Zee, linear-algebra review, pp. 2–4.

Group Theory

The columns of a matrix

For $T : V \to W$, choose bases $(e_1, \ldots, e_n)$ of V and $(w_1, \ldots, w_m)$ of W. Define the $m \times n$ matrix A by

$$T(e_j) = \sum_{i=1}^{m} A_{ij} w_i.$$

Column j records the image of the jth basis vector. Linearity then gives

$$T\left(\sum_j x_j e_j\right) = \sum_i \left(\sum_j A_{ij} x_j\right) w_i, \qquad y_i = \sum_j A_{ij} x_j.$$

Group Theory

Example: the matrix of a reflection

For $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (y, x)$, use the standard basis in both spaces. The basis vectors transform as

$$T(e_1) = T(1, 0) = (0, 1) = e_2, \qquad T(e_2) = T(0, 1) = (1, 0) = e_1.$$

Their coordinate columns give the first and second columns of the matrix:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad X \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}.$$

Group Theory

Matrix multiplication is composition

Apply B first, then A: $x \mapsto Bx \mapsto A(Bx)$. In components,

$$(A(Bx))_i = \sum_j A_{ij} \sum_k B_{jk} x_k = \sum_k \left(\sum_j A_{ij} B_{jk} \right) x_k.$$

Thus the composite matrix is defined by

$$(AB)_{ik} = \sum_j A_{ij} B_{jk}.$$

Multiply row i of A by column k of B. An $m \times n$ matrix times an $n \times p$ matrix gives an $m \times p$ matrix. The rightmost transformation acts first.

Textbook: Zee, linear-algebra review, pp. 3–6.

Group Theory

Exercise: does the order matter?

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

1. Calculate AB and BA.
2. Apply both products to $e_1 = (1, 0)^T$.
3. Does noncommutativity prevent matrix multiplication from being associative?

Keep the order of the transformations explicit: in AB, B acts first.

Group Theory

Exercise: solution

Multiplication gives

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

so $ABe_1 = (2,1)^T$, while $BAe_1 = (1,1)^T$. The transformations do not commute.

No: for any matrices A, B, C of compatible sizes, associativity holds. On every vector,

$$((AB)C)x = A(B(Cx)) = (A(BC))x.$$

Hence $(AB)C = A(BC)$. Two linear maps that agree on every vector have the same matrix; it is enough to check their action on a basis.

Group Theory

Identity, inverse and determinant

The identity matrix $I_n = \mathrm{diag}(1, \ldots, 1)$ leaves every vector fixed. An inverse satisfies $A^{-1}A = AA^{-1} = I_n$.

For a 2×2 matrix, direct multiplication gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc) I_2.$$

Therefore, if $\det A = ad - bc \neq 0$,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, the columns are dependent, so no inverse exists.

Group Theory

The general linear group

The invertible $n \times n$ matrices over $\mathbb{F}$ form the group $\mathrm{GL}(n, \mathbb{F})$:

- Products are invertible: $(AB)^{-1} = B^{-1}A^{-1}$.
- Multiplication is associative.
- The identity is I_n, and inverses are included.

All square matrices together do not form a group under multiplication: singular matrices have no inverse.

We use the standard determinant facts

$$A \text{ invertible} \iff \det A \neq 0, \qquad \det(AB) = \det A \det B.$$

Textbook: Bünemann, §4.1, p. 45; Zee, linear-algebra review, pp. 7–18, 27–29.

Group Theory

The change-of-basis matrix

Let $\mathcal{E}$ be the old basis and $\mathcal{F} = (f_1, \dots, f_n)$ the new one. Put the new basis vectors, in old coordinates, into the columns of S:

$$S = \big([f_1]_{\mathcal{E}} \ \cdots \ [f_n]_{\mathcal{E}}\big).$$

If $x = [v]_{\mathcal{E}}$ and $x' = [v]_{\mathcal{F}}$, then

$$v = \sum_j x'_j f_j \quad \Longrightarrow \quad x = Sx', \qquad x' = S^{-1}x.$$

The columns form a basis, so S is invertible. For $f_1 = (1,1)^T$, $f_2 = (1,-1)^T$,

$$S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad S^{-1} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Group Theory

Why the matrix becomes $S^{-1}AS$

Suppose $y = Ax$ describes a linear map in the old basis. The same input and output have new coordinate columns x', y' with

$$x = Sx', \qquad y = Sy'.$$

Substitution gives

$$Sy' = ASx' \quad \Longrightarrow \quad y' = S^{-1}ASx'.$$

Thus the new matrix is $A' = S^{-1}AS$. Reading right to left: convert to old coordinates, apply A, then convert back to new coordinates. This is a similarity transformation.

Textbook: Zee, linear-algebra review, pp. 23–25; II.1, pp. 93–94.

Group Theory

A reflection in a new basis

For $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, take $f_1 = (1,1)^T$, $f_2 = (1,-1)^T$. Since

$$X f_1 = f_1, \qquad X f_2 = -f_2,$$

its action in the new basis should preserve one coordinate and reverse the other. Indeed,

$$S^{-1} X S = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $v = (3,1)^T = 2f_1 + f_2$, the new coordinates $(2,1)^T$ become $(2,-1)^T$. This gives $2f_1 - f_2 = (1,3)^T$, as expected.

Group Theory

Eigenvectors and eigenvalues

A nonzero vector v is an eigenvector of A if

$$Av = \lambda v.$$

The scalar λ is its eigenvalue. The line spanned by v is preserved, although the vector may change its length, sign or complex phase. We exclude $v = 0$, which would satisfy the equation for every λ.

Rewriting gives $(A - \lambda I)v = 0$. A nonzero solution exists exactly when $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0.$$

Textbook: Zee, linear-algebra review, pp. 20–21.

Group Theory

Finding the eigenvectors: an example

Take $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Its characteristic equation is

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = (\lambda - 3)(\lambda - 1) = 0.$$

- For $\lambda = 3$, $(A - 3I)v = 0$ gives $-x + y = 0$. Choose $v_+ = (1, 1)^T$.
- For $\lambda = 1$, $(A - I)v = 0$ gives $x + y = 0$. Choose $v_- = (1, -1)^T$.

The eigenvectors are independent. Any nonzero multiple of either vector is another eigenvector with the same eigenvalue.

Group Theory

Diagonalization

Suppose A has a basis of eigenvectors $v_1, \ldots, v_n$, with eigenvalues $\lambda_1, \ldots, \lambda_n$. Put these vectors into the columns of S. Then

$$AS = (Av_1 \ \cdots \ Av_n) = (\lambda_1 v_1 \ \cdots \ \lambda_n v_n) = S \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Multiplying on the left by S^{-1} gives

$$S^{-1}AS = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

For the previous example, $S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ gives $S^{-1}AS = \operatorname{diag}(3, 1)$. Independence matters: it guarantees that S^{-1} exists.

Textbook: Zee, linear-algebra review, pp. 23–24.

Group Theory

Not every matrix is diagonalizable

Consider

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \det(J - \lambda I) = (1 - \lambda)^2.$$

The only eigenvalue is 1. But

$$(J - I)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix} = 0 \quad \Longrightarrow \quad y = 0.$$

All eigenvectors lie on a single line, so they cannot form a basis of a two-dimensional space. Thus J is not diagonalizable.

A repeated eigenvalue is not itself a problem: I_2 is diagonal and every nonzero vector is an eigenvector. The condition is a full basis of eigenvectors.

Group Theory

Trace and cyclicity

The trace is the sum of diagonal entries: $\operatorname{tr} A = \sum_i A_{ii}$. Although AB need not equal BA, their traces agree:

$$\operatorname{tr}(AB) = \sum_{i,j} A_{ij} B_{ji} = \sum_{j,i} B_{ji} A_{ij} = \operatorname{tr}(BA).$$

Here the scalar entries commute. Applying $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ with $X = AB$, $Y = C$, and then with $X = CA$, $Y = B$, gives

$$\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}(BCA).$$

Textbook: Zee, linear-algebra review, pp. 6–7.

Group Theory

The trace does not depend on the basis

Under a similarity transformation,

$$\operatorname{tr}(S^{-1}AS) = \operatorname{tr}(ASS^{-1}) = \operatorname{tr} A.$$

If A has an eigenbasis, its diagonal form immediately gives

$$\operatorname{tr} A = \sum_i \lambda_i, \qquad \det A = \prod_i \lambda_i.$$

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the trace is $4 = 3 + 1$ and the determinant is $3 = 3 \cdot 1$.

In Lecture 8, the trace of a representation matrix will connect this basis independence to the conjugacy classes of Week 3.

Textbook: Zee, linear-algebra review, p. 25; II.1, pp. 93–94.

Group Theory

Complex conjugation and the adjoint

For $z = a + ib$, define $z^* = a - ib$, so $z^* z = a^2 + b^2$. The transpose interchanges rows and columns, $(A^T)_{ij} = A_{ji}$. The adjoint also conjugates the entries:

$$A^\dagger = (A^*)^T, \qquad \begin{pmatrix} 1 & i \\ 2 & 3-i \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 2 \\ -i & 3+i \end{pmatrix}.$$

A column v becomes a row $v^\dagger$. Transposition reverses the order of a product, and conjugation preserves multiplication of scalars, so

$$(AB)^\dagger = B^\dagger A^\dagger.$$

Textbook: Zee, linear-algebra review, pp. 6 and 19.

Group Theory

Inner products and orthonormal vectors

For complex vectors, the standard inner product and squared norm are

$$\langle u, v\rangle = u^\dagger v = \sum_i u_i^* v_i, \qquad \|v\|^2 = v^\dagger v = \sum_i |v_i|^2.$$

Conjugation is essential: for $v = (1, i)^T$, $v^T v = 0$ but $v^\dagger v = 2$.

Vectors are orthogonal if $u^\dagger v = 0$, and normalized if $v^\dagger v = 1$. An orthonormal basis satisfies $u_i^\dagger u_j = \delta_{ij}$, where δ_{ij} is 1 for $i = j$ and 0 otherwise. For example,

$$u_1 = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad u_2 = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Group Theory

Orthogonal and unitary matrices

A real square matrix O is orthogonal if $O^T O = I$. A complex square matrix U is unitary if $U^\dagger U = I$. Therefore $O^{-1} = O^T$ and $U^{-1} = U^\dagger$.

Their columns are orthonormal. They preserve inner products:

$$(Uu)^\dagger (Uv) = u^\dagger U^\dagger U v = u^\dagger v.$$

Rotations and reflections are examples:

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Both are orthogonal; their determinants are $+1$ and -1, respectively.

Textbook: Zee, linear-algebra review, pp. 24–25.

Group Theory

Hermitian matrices

A matrix is Hermitian if $H^\dagger = H$. If $Hv = \lambda v$ with $v \neq 0$, then

$$v^\dagger H v = \lambda v^\dagger v.$$

Using the Hermitian condition explicitly,

$$v^\dagger H v = v^\dagger H^\dagger v = (Hv)^\dagger v = \lambda^* v^\dagger v.$$

Thus $(\lambda - \lambda^*)v^\dagger v = 0$. Since $v^\dagger v > 0$, λ is real.

Hermitian and unitary are different conditions: $\mathrm{diag}(2, 3)$ is Hermitian but not unitary.

Textbook: Zee, linear-algebra review, pp. 19–25.

Group Theory

Hermitian matrices: orthogonal eigenvectors

Let $Hu = \lambda u$ and $Hv = \mu v$. The eigenvalues are real, so

$$\mu u^\dagger v = u^\dagger H v = u^\dagger H^\dagger v = (Hu)^\dagger v = \lambda u^\dagger v.$$

Hence $(\mu - \lambda) u^\dagger v = 0$. If $\lambda \neq \mu$, then $u^\dagger v = 0$: eigenvectors with distinct eigenvalues are orthogonal.

We use the spectral theorem: every Hermitian matrix has an orthonormal eigenbasis. Thus $U^\dagger H U$ is real diagonal for some unitary U. An orthonormal basis can also be chosen within each degenerate eigenspace.

Textbook: Zee, linear-algebra review, pp. 19–25.

Group Theory

Invariant subspaces

A subspace W is invariant under a linear map T if $T(w) \in W$ for every $w \in W$. Vectors stay in W; they need not stay fixed.

An eigenvector spans an invariant line. If a basis begins with a basis of W, the matrix has the form

$$A = \begin{pmatrix} B & C \\ 0 & K \end{pmatrix}, \qquad A \begin{pmatrix} w \\ 0 \end{pmatrix} = \begin{pmatrix} Bw \\ 0 \end{pmatrix}.$$

The lower-left block vanishes because inputs in W cannot leave W. If the remaining basis vectors also span an invariant subspace, the upper-right block vanishes too: the matrix is block diagonal.

Textbook: Bünemann, pp. 47–49; Zee, II.1, pp. 94–96.

Group Theory

Check your understanding

Consider the plane

$$W = \{(x, y, z)^T : x + y + z = 0\} \subseteq \mathbb{R}^3.$$

1. Show that W is a subspace.
2. Show that $a = (1, -1, 0)^T$ and $b = (0, 1, -1)^T$ form a basis.
3. Express $(2, -1, -1)^T$ in this basis. What is $\dim W$?

This plane will reappear in the permutation representation of S_3.

Group Theory

Check your understanding: solution

The zero vector lies in W. Linear combinations of zero-sum vectors still have zero component sum, so W is a subspace.

$$\alpha a + \beta b = (\alpha, -\alpha + \beta, -\beta)^T.$$

For $(x, y, z)^T \in W$, choose $\alpha = x$, $\beta = -z$. The middle component is $-x - z = y$, proving spanning. If the combination is zero, its first and third components give $\alpha = \beta = 0$, proving independence.

Thus $\dim W = 2$, and

$$(2, -1, -1)^T = 2a + b.$$

Group Theory

Lecture 7 in brief

- A basis gives unique coordinates. A matrix records a linear map through the images of the basis vectors.
- Composition is matrix multiplication. The rightmost matrix acts first.
- Changing basis on both input and output gives $A' = S^{-1}AS$. An eigenbasis makes A' diagonal; the trace is unchanged.
- Unitary matrices preserve inner products. An invariant subspace is preserved as a set, not necessarily pointwise.

Next: represent each group element by a matrix, then look for subspaces preserved by the whole group.

Textbook: Zee, linear-algebra review, pp. 1–8, 19–25; Bünemann, §4.1.

Representations and Invariant Subspaces

Week 4 · Lecture 8

Mehmet Özkan

Group Theory

Reading for Lecture 8

- **Zee, II.1, pp. 89–95:** representations, faithfulness, characters, equivalence and direct sums.
- **Bünemann, §4.1.1, pp. 45–48, and §4.2, pp. 50–53:** equivalent and irreducible representations; the D_3 example.
- **Zee, II.1, pp. 95–97; Bünemann, pp. 46–49:** unitarity and invariant complements. The detailed unitarity proof and subgroup restriction at the end are optional.

We use $r = (123)$, $s = (23)$ and right-to-left composition. Representations are finite dimensional over $\mathbb{C}$ unless a real space is explicitly specified. The groups in this lecture are finite.

Group Theory

What is a representation?

A d-dimensional representation of G is a homomorphism

$$D : G \longrightarrow \mathrm{GL}(d, \mathbb{C}), \qquad D(gh) = D(g)D(h).$$

A matrix $D(g)$ represents each element g. Multiplication in G becomes matrix multiplication. Equivalently, G acts by invertible linear maps on a vector space of dimension d.

The order $|G|$ counts group elements; the dimension d is the size of the matrices. A group with six elements may have representations by 1×1, 2×2 or larger matrices. Different elements may have the same matrix.

Textbook: Zee, II.1, pp. 89–92; Bünemann, §4.2, pp. 50–51.

Group Theory

The identity and inverses

Since $eg = g$, the representation law gives

$$D(e)D(g) = D(g).$$

The matrix $D(g)$ is invertible because the codomain is $\mathrm{GL}(d, \mathbb{C})$. Multiplying on the right by $D(g)^{-1}$ gives $D(e) = I_d$.

Likewise,

$$D(g)D(g^{-1}) = D(gg^{-1}) = D(e) = I_d,$$

and the reverse product is also I_d. Therefore

$$D(g^{-1}) = D(g)^{-1}.$$

The group identity e and the identity matrix I_d are different objects.

Group Theory

Two one-dimensional representations of S_3

The trivial representation assigns

$$D_{\mathrm{triv}}(g) = 1 \qquad \text{for every } g \in S_3.$$

It preserves products because $1 \cdot 1 = 1$.

The sign of a permutation gives another representation:

$$D_{\mathrm{sgn}}(g) = \mathrm{sgn}(g), \qquad \mathrm{sgn}(gh) = \mathrm{sgn}(g)\,\mathrm{sgn}(h).$$

The even permutations e, r, r^2 map to $+1$; the transpositions $s, rs, r^2 s$ map to -1. Both assignments are valid, although neither distinguishes all six group elements.

Textbook: Zee, II.1, p. 92; Bünemann, §4.2, pp. 51–52.

Group Theory

Faithful representations and kernels

A representation is faithful if $D(g) = D(h)$ implies $g = h$. Its kernel is

$$\ker D = \{g \in G : D(g) = I_d\}.$$

As for any group homomorphism,

$$D \text{ faithful} \quad \Longleftrightarrow \quad \ker D = \{e\}.$$

Indeed, $D(g) = D(h)$ implies $D(h^{-1}g) = I_d$. A trivial kernel then gives $h^{-1}g = e$, so $g = h$. The reverse implication follows from injectivity and $D(e) = I_d$.

For the sign representation, $\ker D_{\text{sgn}} = A_3 = \{e, r, r^2\}$. The first isomorphism theorem gives $S_3/A_3 \cong \{+1, -1\}$.

Group Theory

The three one-dimensional representations of $\mathbb{Z}_3$

Let $r^3 = e$ and $\omega = e^{2\pi i/3}$. A one-dimensional representation must satisfy $D(r)^3 = 1$. The possibilities are

	e	r	r^2
D_0	1	1	1
D_1	1	ω	ω^2
D_2	1	ω^2	ω

They preserve multiplication because powers add modulo three. The last two are faithful; D_0 is trivial.

Although D_1 and D_2 have the same set of values, they assign different values to the same group element r.

Textbook: Zee, II.1, p. 92; Bünemann, §4.2, p. 51.

Group Theory

The permutation representation of S_3

Use the standard basis e_1, e_2, e_3 of $\mathbb{C}^3$ and define

$$P(p)e_j = e_{p(j)}.$$

Column j has its 1 in row $p(j)$, with zeros elsewhere. For $r = (123)$ and $s = (23)$,

$$P(r) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad P(s) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

For example, the second column of $P(r)$ is e_3 because $r(2) = 3$. Each matrix is orthogonal: its columns are a rearrangement of an orthonormal basis.

Textbook: Zee, II.1, pp. 90–91, gives the analogous construction for S_4.

Group Theory

Why permutation matrices form a representation

For every basis vector,

$$P(p)P(q)e_j = P(p)e_{q(j)} = e_{p(q(j))} = P(pq)e_j.$$

Equality on a basis gives $P(p)P(q) = P(pq)$. For instance, $rs = (123)(23) = (12)$ and

$$P(r)P(s) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = P((12)).$$

The representation is faithful: if $P(p) = P(q)$, their action on every e_j gives $p(j) = q(j)$, so $p = q$.

Group Theory

Basis labels and vector components

The cycle $r = (123)$ sends $e_1 \to e_2 \to e_3 \to e_1$. Therefore

$$P(r)(xe_1 + ye_2 + ze_3) = xe_2 + ye_3 + ze_1,$$

which in the original ordered basis has coordinate column

$$P(r)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} z \\ x \\ y \end{pmatrix}.$$

The coefficient x travels with e_1 to the e_2 position. In general, $(P(p)v)_i = v_{p^{-1}(i)}$.

This is the same distinction as in Week 3: the destination of a label and the contents of a fixed position are different descriptions.

Group Theory

The plane representation of D_3

Place the triangle's vertices at $(1, 0)$, $(-1/2, \sqrt{3}/2)$ and $(-1/2, -\sqrt{3}/2)$, labelled $1, 2, 3$. Then $r = (123)$ is a $120°$ counterclockwise rotation; $s = (23)$ is reflection in the horizontal axis through vertex 1:

$$D(r) = R = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad D(s) = F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices act on a two-dimensional plane. The permutation representation acted on a three-dimensional space with one basis vector per vertex. They represent the same group in different dimensions.

Textbook: Bünemann, §4.2, p. 52. His complex matrices are obtained by the basis change on the optional slides.

Group Theory

Checking the D_3 relations

Direct multiplication gives

$$R^2 = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad R^3 = I_2, \qquad F^2 = I_2.$$

Multiplication by F changes the sign of the second row on the left and the second column on the right. Hence

$$FRF = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix} = R^2 = R^{-1}.$$

The three rotations are distinct, as are their products with F. Their determinants are $+1$ and -1, respectively. Thus all six matrices are distinct: this representation is faithful.

Group Theory

Why the relations give a representation

The six pairs (a, b), $a = 0, 1, 2$, $b = 0, 1$, give the six elements of S_3, each uniquely as $r^a s^b$. Thus $D(r^a s^b) = R^a F^b$ is well defined. From $sr = r^{-1}s$ and $FR = R^{-1}F$,

$$s^b r^c = r^{(-1)^b c} s^b, \qquad F^b R^c = R^{(-1)^b c} F^b.$$

Consequently, the products follow exactly the same rule:

$$(r^a s^b)(r^c s^d) = r^{a+(-1)^b c} s^{b+d},$$
$$(R^a F^b)(R^c F^d) = R^{a+(-1)^b c} F^{b+d}.$$

Since $r^3 = e$, $s^2 = e$ and $R^3 = F^2 = I$, both rules reduce exponents modulo three and two. Hence $D(gh) = D(g)D(h)$.

Group Theory

Equivalent representations

Representations D and D' are equivalent if one invertible matrix S satisfies

$$D'(g) = S^{-1}D(g)S \qquad \text{for every } g \in G.$$

The same basis change must work for the whole group. It preserves the representation law:

$$D'(g)D'(h) = S^{-1}D(g)\underbrace{SS^{-1}}_{I}D(h)S = S^{-1}D(gh)S = D'(gh).$$

In one dimension, S is a nonzero scalar and cancels. Thus two one-dimensional representations are equivalent only if they agree on every element: D_1 and D_2 of $\mathbb{Z}_3$ are inequivalent.

Textbook: Zee, II.1, pp. 93–94; Bünemann, pp. 45–46, 51.

Group Theory

Equivalent does not mean visibly orthogonal

For the plane representation, change basis with $S = \mathrm{diag}(2, 1)$. Then

$$R' = S^{-1}RS = \begin{pmatrix} -1/2 & -\sqrt{3}/4 \\ \sqrt{3} & -1/2 \end{pmatrix}, \qquad F' = S^{-1}FS = F.$$

These matrices satisfy the same group relations. But the first column of R' has squared length

$$\left(-\frac{1}{2}\right)^2 + (\sqrt{3})^2 = \frac{13}{4} \neq 1,$$

so R' is not orthogonal in the standard coordinate inner product. It is still a valid representation, equivalent to an orthogonal one. A nonorthonormal basis can hide length preservation.

Group Theory

Reducible and irreducible representations

A subspace W is invariant under D if

$$D(g)w \in W \qquad \text{for every } g \in G,\ w \in W.$$

The representation is reducible if a nonzero proper invariant subspace exists: $\{0\} \subsetneq W \subsetneq V$. Otherwise it is irreducible.

The same W must work for every group element. For D_3, checking the generators suffices: products preserve W, and $r^{-1} = r^2$, $s^{-1} = s$.

Every one-dimensional representation is irreducible, including the trivial one: a one-dimensional space has no nonzero proper subspace.

Textbook: Zee, II.1, pp. 94–96; Bünemann, pp. 47–49.

Group Theory

Direct sums

The notation $V = W \oplus U$ means that every vector has a unique decomposition $v = w + u$. Equivalently, $V = W + U$ and $W \cap U = \{0\}$.

If both subspaces are invariant, a basis formed from their two bases gives

$$D(g) = \begin{pmatrix} D_W(g) & 0 \\ 0 & D_U(g) \end{pmatrix} = D_W(g) \oplus D_U(g).$$

Each block obeys the representation law, and $\dim V = \dim W + \dim U$. The goal of reduction is to find one basis that puts every representation matrix into such a block form.

Textbook: Zee, II.1, pp. 94–95; Bünemann, pp. 47, 53.

Group Theory

Why unitarity supplies an invariant complement

For an invariant subspace W, define

$$W^{\perp} = \{u : w^{\dagger}u = 0 \text{ for every } w \in W\}.$$

In a finite-dimensional inner-product space, $V = W \oplus W^{\perp}$. Suppose every $D(g)$ is unitary. For $u \in W^{\perp}$, $w \in W$,

$$w^{\dagger}D(g)u = \big(D(g)^{\dagger}w\big)^{\dagger}u = \big(D(g^{-1})w\big)^{\dagger}u = 0.$$

The last equality holds because $D(g^{-1})w \in W$. Thus $W^{\perp}$ is invariant too, and a basis adapted to the two spaces makes all matrices block diagonal.

Textbook: Bünemann, pp. 48–49.

Group Theory

The finite-group unitarity theorem

Every finite-dimensional complex representation of a finite group is equivalent to a unitary representation. Over $\mathbb{R}$, it can be made orthogonal.

The theorem asserts that a suitable common basis exists; it does not assert that every original matrix is already unitary. An optional proof appears at the end.

Combining this theorem with the invariant-complement argument, every finite-group representation over $\mathbb{R}$ or $\mathbb{C}$ is a direct sum of irreducible pieces. Repeated splitting must stop because the dimensions decrease.

Textbook: Zee, II.1, pp. 96–97; Bünemann, §4.1.1, pp. 46–49.

Group Theory

The permutation representation is reducible

Every permutation fixes $(1,1,1)^T$. Hence the line

$$W_0 = \mathrm{span}\{(1,1,1)^T\}$$

is invariant, and every group element acts as 1 on it.

Its orthogonal complement is

$$W_2 = \{(x,y,z)^T : x+y+z=0\}.$$

Permutations preserve the sum of the entries, so W_2 is invariant as well. Therefore $\mathbb{C}^3 = W_0 \oplus W_2$. We now choose a basis that exhibits this decomposition explicitly.

Group Theory

A basis adapted to the two invariant spaces

Choose

$$u_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad u_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \qquad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

The vector u_0 spans W_0; the vectors u_1, u_2 lie in W_2. Their pairwise dot products vanish and their squared norms are one. Thus they form an orthonormal basis.

Set $Q = (u_0\ u_1\ u_2)$. Then $Q^{-1} = Q^T$, and the matrices in the new basis are $Q^T P(g) Q$. Column j records the image of the corresponding new basis vector.

Group Theory

The rotation in the adapted basis

Since $P(r)(x,y,z)^T = (z,x,y)^T$,

$$P(r)u_0 = u_0, \qquad P(r)u_1 = \tfrac{1}{\sqrt{6}}(-1,2,-1)^T.$$

Taking dot products gives $u_1^T P(r) u_1 = -1/2$ and $u_2^T P(r) u_1 = \sqrt{3}/2$. Similarly,

$$P(r)u_2 = -\tfrac{\sqrt{3}}{2}u_1 - \tfrac{1}{2}u_2.$$

These coefficients form the columns:

$$Q^T P(r) Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & -1/2 \end{pmatrix} = 1 \oplus R.$$

Group Theory

The reflection and the decomposition

The permutation $s = (23)$ exchanges the second and third components. Therefore

$$P(s)u_0 = u_0, \qquad P(s)u_1 = u_1, \qquad P(s)u_2 = -u_2.$$

Hence

$$Q^T P(s) Q = \mathrm{diag}(1, 1, -1) = 1 \oplus F.$$

Since r, s generate S_3, the same Q separates every matrix:

$$P \simeq D_{\mathrm{triv}} \oplus D_{\mathrm{plane}}, \qquad 3 = 1 + 2.$$

The numbers denote dimensions. We have found the two blocks explicitly.

Group Theory

Why these blocks are exactly the triangle matrices

Project each standard basis vector e_j onto the zero-sum plane:

$$p_j = e_j - (u_0^T e_j)u_0 = e_j - \tfrac{1}{3}(1,1,1)^T.$$

In the orthonormal basis (u_1, u_2), their coordinates are

$$[p_1] = \sqrt{\tfrac{2}{3}}\begin{pmatrix}1\\0\end{pmatrix}, \quad [p_2] = \sqrt{\tfrac{2}{3}}\begin{pmatrix}-1/2\\\sqrt{3}/2\end{pmatrix}, \quad [p_3] = \sqrt{\tfrac{2}{3}}\begin{pmatrix}-1/2\\-\sqrt{3}/2\end{pmatrix}.$$

These are the labelled triangle vertices, all scaled by $\sqrt{2/3}$. Our choice of u_1 points towards vertex 1:

$$p_1 = \sqrt{\tfrac{2}{3}}\, u_1.$$

Group Theory

Why the vertex permutation fixes the matrix

Since $P(g)$ fixes $(1,1,1)^T$, we have

$$P(g)p_j = P(g)(e_j - \tfrac{1}{3}(1,1,1)^T) = e_{g(j)} - \tfrac{1}{3}(1,1,1)^T = p_{g(j)}.$$

This is the same vertex permutation as in the triangle picture. The vectors p_1, p_2 are independent and span the plane, so agreeing on these two vectors fixes the linear map.

Thus r and s act in the (u_1, u_2) coordinates by exactly R and F. This uses Lecture 7's rule: it is enough to check a linear map on a basis.

We still need to check that the two-dimensional block is irreducible.

Group Theory

Why the plane representation is irreducible

In two dimensions, a proper nonzero subspace is a line. An invariant line of $F = \mathrm{diag}(1, -1)$ must be an eigenline: one of the coordinate axes.

But neither axis is preserved by R:

$$R\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}-1/2\\ \sqrt{3}/2\end{pmatrix}, \qquad R\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-\sqrt{3}/2\\ -1/2\end{pmatrix}.$$

Thus no line is invariant under both generators. The representation is irreducible over both $\mathbb{R}$ and $\mathbb{C}$.

Diagonalizing one generator does not reduce a representation: the decomposition must be preserved by the whole group.

Textbook: Bünemann, §4.2, p. 52, identifies this irreducible representation.

Group Theory

Faithfulness and irreducibility are different

For the representations of S_3 constructed so far:

Representation	Dimension	Faithful?	Irreducible?
Trivial	1	No	Yes
Sign	1	No	Yes
Plane	2	Yes	Yes
Permutation	3	Yes	No

Faithfulness distinguishes group elements. Irreducibility excludes proper nonzero invariant subspaces.

These three irreducible representations are inequivalent. Character theory will show that the list is complete.

Group Theory

The character of a representation

Define the character by

$$\chi_D(g) = \operatorname{tr} D(g).$$

At the identity, $\chi_D(e) = \operatorname{tr} I_d = d$: the character records the dimension.

Equivalent representations have the same character because

$$\operatorname{tr}(S^{-1} D(g) S) = \operatorname{tr} D(g).$$

A difference in characters at even one element proves inequivalence. Also,

$$\chi_{D \oplus D'}(g) = \chi_D(g) + \chi_{D'}(g),$$

since the trace of a block-diagonal matrix is the sum of the block traces.

Textbook: Zee, II.1, pp. 93–95; Bünemann, §4.2, pp. 52–53.

Group Theory

Character is a function of class

If $a' = h^{-1}ah$, the representation law gives

$$D(a') = D(h)^{-1}D(a)D(h).$$

Taking the trace,

$$\chi_D(a') = \operatorname{tr}\left(D(h)^{-1}D(a)D(h)\right) = \operatorname{tr} D(a) = \chi_D(a).$$

Thus conjugate elements have the same character. The matrices themselves may differ.

Here conjugation compares two elements within one representation. Equivalence compares two representations of the same element, using a fixed change of basis S.

Textbook: Zee, II.1, p. 93; Bünemann, §4.2, pp. 52–53.

Group Theory

Checking the decomposition with traces

A permutation matrix has a 1 on the diagonal exactly at a fixed label, so its trace counts fixed labels. For the three conjugacy classes,

	$\{e\}$	$\{r, r^2\}$	$\{s, rs, r^2 s\}$
χ_P	3	0	1
χ_{triv}	1	1	1
χ_{plane}	2	-1	0

Directly, $\operatorname{tr} I_2 = 2$, $\operatorname{tr} R = -1$, $\operatorname{tr} F = 0$. Column by column,

$$\chi_P = \chi_{\text{triv}} + \chi_{\text{plane}}.$$

This agrees with our explicit decomposition.

Group Theory

Check your understanding: representations of $\mathbb{Z}_4$

Let $\mathbb{Z}_4 = \langle a \mid a^4 = e \rangle$ and define

$$D_k(a^m) = \exp(2\pi i k m/4), \qquad k = 0, 1, 2, 3.$$

1. Check that each D_k preserves multiplication and is well defined modulo four.
2. Find the kernel of each representation. Which are faithful?
3. Which are irreducible over $\mathbb{C}$?

The kernel condition is $km \equiv 0 \pmod 4$.

Group Theory

Check your understanding: solution

The exponents add under multiplication. Increasing m by four multiplies the value by $e^{2\pi ik} = 1$, so the definition is well defined.

k	Kernel	Faithful?
0	$\{e, a, a^2, a^3\}$	No
1	$\{e\}$	Yes
2	$\{e, a^2\}$	No
3	$\{e\}$	Yes

All four are irreducible because their vector spaces have dimension one. A nontrivial kernel does not imply reducibility.

Group Theory

Optional: restriction to the rotation subgroup

Restrict the plane representation from D_3 to $\langle r \rangle \cong \mathbb{Z}_3$. Over $\mathbb{R}$, it remains irreducible: $\det(R - \lambda I) = \lambda^2 + \lambda + 1$ has no real roots, so there is no real invariant line.

Over $\mathbb{C}$, take $\omega = e^{2\pi i/3}$ and

$$v_+ = \begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad v_- = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Then $Rv_+ = \omega v_+$ and $Rv_- = \omega^2 v_-$. With $S = (v_+ \; v_-)$,

$$S^{-1} R S = \operatorname{diag}(\omega, \omega^2).$$

The complex representation of the subgroup splits as $D_1 \oplus D_2$.

Textbook: Zee, II.1, pp. 92, 95–96.

Group Theory

Optional: Bünemann's complex basis

In the same basis, the reflection exchanges v_+ and v_-:

$$Fv_+ = v_-, \qquad Fv_- = v_+, \qquad S^{-1}FS = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Since $\omega \neq \omega^2$, these are the only eigenlines of R, hence its only invariant lines. Neither is preserved by F, so the full D_3 representation is irreducible over $\mathbb{C}$.

The matrices $\mathrm{diag}(\omega, \omega^2)$ and $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ are precisely Bünemann's $\widetilde{D}_2$ and $\widetilde{D}_4$. Their squares and products give his other four matrices. His complex matrices and our real geometric matrices are equivalent.

Textbook: Bünemann, Eq. (2.16), p. 26, and Eq. (4.14), p. 52.

Group Theory

Optional: an invariant positive inner product

For a representation of a finite group, define the Hermitian matrix

$$H = \sum_{g \in G} D(g)^\dagger D(g).$$

For $v \neq 0$, $v^\dagger H v = \sum_g \|D(g)v\|^2 > 0$: the identity term alone contributes $\|v\|^2 > 0$. Thus H is positive definite.

The map $g \mapsto gh$ permutes the group elements, so

$$D(h)^\dagger H D(h) = \sum_g D(gh)^\dagger D(gh) = H.$$

Therefore $u^\dagger H v$ is an inner product preserved by every group element. We now choose coordinates in which it becomes the usual inner product.

Group Theory

Optional: making every matrix unitary

The spectral theorem gives $H = U_H \,\mathrm{diag}(h_1, \ldots, h_d) U_H^\dagger$ with U_H unitary and $h_i > 0$. Set

$$C = \mathrm{diag}(\sqrt{h_1}, \ldots, \sqrt{h_d}) U_H^\dagger, \qquad H = C^\dagger C.$$

Since $h_i > 0$, C is invertible. Set $D_{\mathrm{u}}(g) = C D(g) C^{-1}$: this is equivalence with $S = C^{-1}$. Then

$$\begin{aligned} D_{\mathrm{u}}(g)^\dagger D_{\mathrm{u}}(g) &= (C^{-1})^\dagger D(g)^\dagger H D(g) C^{-1} \\ &= (C^{-1})^\dagger H C^{-1} = I_d. \end{aligned}$$

The same C makes every matrix unitary. This completes the finite-group unitarity proof.

Textbook: Zee, II.1, pp. 96–97; compare Bünemann, pp. 46–47.

Group Theory

Week 4 in brief

- A representation is a homomorphism into $\mathrm{GL}(d, \mathbb{C})$. It is faithful exactly when its kernel is trivial.
- Equivalent representations use one common basis change, $D'(g) = S^{-1}D(g)S$.
- Reduction separates invariant subspaces. For S_3, the permutation representation decomposes as $1 \oplus D_{\mathrm{plane}}$, with the two-dimensional piece irreducible.
- Characters are traces, add under direct sums and are constant on conjugacy classes.

Next week: Schur's lemma, orthogonality and characters.

Textbook: Zee, II.1, pp. 89–97; Bünemann, §4.1 and §4.2, pp. 45–53.